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Linear Regression Explained

Salary Example

Every ML engineer must nail this. Here's the visual way.

Don't just memorize formulas. Learn how it visually maps inputs to outputs, makes predictions, and optimizes its errors.

- Regression
- Salary Predictor
- Gradient Descent
- Visual Guide

→ **Swipe through.** Let's break down how this model learns the line.



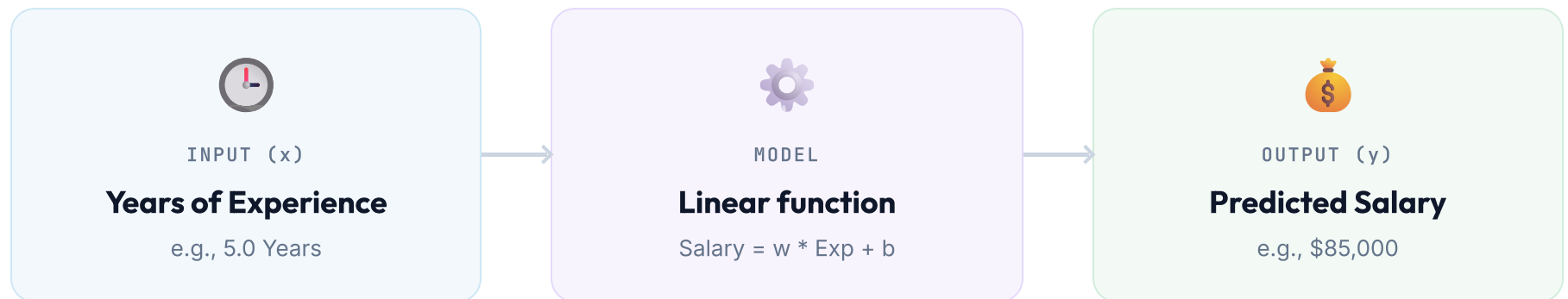
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Predicting Continuous Output

Linear Regression predicts a **continuous numerical value** (e.g., salary, price, height) rather than classifying into categories.



Key Concept: The model looks at historical data points, learns the linear trend (correlation), and projects it to make predictions on unseen variables.

✓ Continuous Value

✓ Linear Dependent



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Deconstructing the Math

Don't let the symbols intimidate you. It's just a formula for a straight line:

y

SALARY

=

w

WEIGHT (SLOPE)

×

X

EXPERIENCE

+

b

BIAS (Y-INTERCEPT)

- w

Weight / Slope (w)
How much salary increases per year of experience. If **w = \$10,000**, then each year adds \$10k to your paycheck.
- b

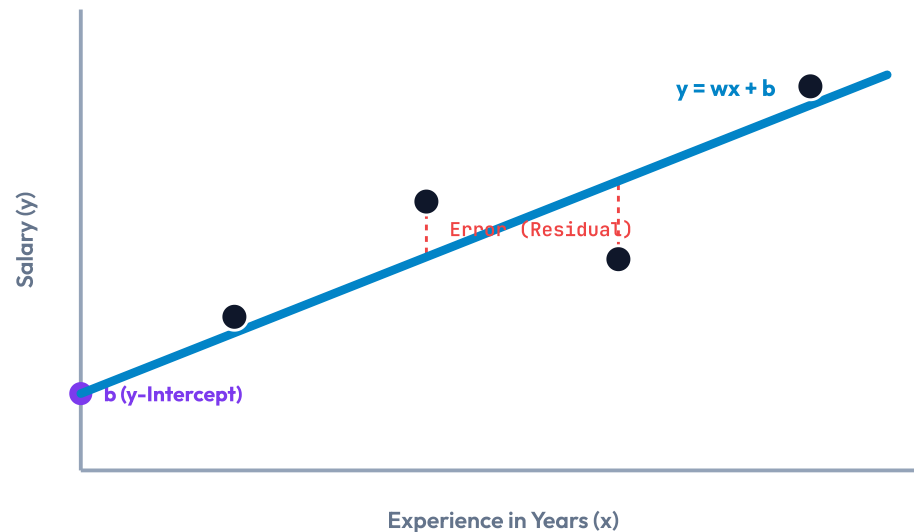
Bias / Intercept (b)
The starting base salary. The prediction when experience is **0 years**. If **b = \$50,000**, that's your baseline starting salary.

Example: Salary = \$10k * Experience + \$50k



Fitting the Best Straight Line

The model finds the line that goes as close as possible to all data points.



● **Residual (Error):** The vertical distance between the actual salary (black dot) and the predicted salary on the line. The model adjusts the line to minimize these squared gaps.

Goal: Minimize Residual Sum of Squares (RSS)



Gradient Descent Intuition

How does the model find the best w and b ? It uses **Gradient Descent** — taking small steps downhill to reach the lowest possible error.



1 Predict & Score

Calculate salary predictions and measure error gap (Loss).

2 Adjust Downhill

Tweak w and b slightly in the direction of lower error.

Iterates automatically until the line stabilizes



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When to Choose Linear Regression

Don't reach for complex neural networks first. Use Linear Regression when:



Relationship is Linear

The output changes proportionally with the input. If Years of Experience doubles, salary generally shows a steady upward trend.



Predicting Numeric Output

When you need a continuous numerical output (Salary, Stock Price, Temp) rather than category labels (Spam vs. Not Spam).



Interpretability is Key

You need to explain exactly how every input affects the prediction. Perfect for auditability and establishing a baseline.



Tip: Always use it as your baseline regression model.



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💬 What was your first ML model?

Linear regression is simple, but it teaches you the fundamentals of **parameters**, **loss functions**, and **optimization**.

What regression model do you use as your baseline?

Let me know in the comments below! 🙌

Day 11 Recap:

🎯 Predicts continuous outputs

📊 Line equation: $y = wx + b$

🔧 w is slope, b is y-intercept

⚠️ Least Squares minimizes error

💡 Gradient Descent updates weights

🚀 Quick to train, easy to interpret

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#DataScience

#LinearRegression

#AIEducation

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